

✓ ZoyaSofts Internship Training

Week 1 — AI & Python Foundations

Day 6 — Mini Project

AI Rule-Based Recommendation System

Project Objective

Build a simple rule-based recommendation system that takes user preferences as input, applies predefined rules, and produces a recommendation.

Project Flow

User Preferences ↓ Rule Engine ↓ Recommendation

Inputs

The system will take:

- Interest area
- Skill level

Output

The system will recommend a suitable learning track based on the user's preferences.

```
# AI Rule-Based Recommendation System

def recommend_course(interest, level):
    """
    Recommend a learning track based on user interest and skill level.
    """

    # Rule 1: AI/ML + Beginner
    if interest == "AI/ML" and level == "Beginner":
        return "AI/ML Foundations – Beginner Track"

    # Rule 2: AI/ML + Advanced
    elif interest == "AI/ML" and level == "Advanced":
        return "AI/ML Internship – Advanced Track"

    # Rule 3: Web Dev + Intermediate
    elif interest == "Web Dev" and level == "Intermediate":
        return "Python Full Stack – Intermediate Track"

    # Rule 4: Data Analysis + Beginner
    elif interest == "Data Analysis" and level == "Beginner":
        return "Excel + Data Analysis – Beginner Track"

    # Rule 5: Web Dev + Beginner
    elif interest == "Web Dev" and level == "Beginner":
        return "Web Development – Beginner Track"

    # Rule 6: Data Analysis + Advanced
```

```
elif interest == "Data Analysis" and level == "Advanced":
    return "Python Data Analysis – Advanced Track"

# Default rule for unmatched combinations
else:
    return "No exact recommendation found. Please try another combination."

print("Recommendation System Ready!")
```

Recommendation System Ready!

```
# Test 1
print("Test 1:")
print("Interest: AI/ML")
print("Level: Beginner")
print("Recommendation:", recommend_course("AI/ML", "Beginner"))

# Test 2
print("\nTest 2:")
print("Interest: AI/ML")
print("Level: Advanced")
print("Recommendation:", recommend_course("AI/ML", "Advanced"))

# Test 3
print("\nTest 3:")
print("Interest: Web Dev")
print("Level: Intermediate")
print("Recommendation:", recommend_course("Web Dev", "Intermediate"))

# Test 4
print("\nTest 4:")
print("Interest: Data Analysis")
print("Level: Beginner")
print("Recommendation:", recommend_course("Data Analysis", "Beginner"))

# Test 5
print("\nTest 5:")
print("Interest: Web Dev")
print("Level: Advanced")
print("Recommendation:", recommend_course("Web Dev", "Advanced"))
```

Test 1:
Interest: AI/ML
Level: Beginner
Recommendation: AI/ML Foundations – Beginner Track

Test 2:
Interest: AI/ML
Level: Advanced
Recommendation: AI/ML Internship – Advanced Track

Test 3:
Interest: Web Dev
Level: Intermediate
Recommendation: Python Full Stack – Intermediate Track

Test 4:
Interest: Data Analysis
Level: Beginner
Recommendation: Excel + Data Analysis – Beginner Track

Test 5:
Interest: Web Dev
Level: Advanced
Recommendation: No exact recommendation found. Please try another combination.

Project Explanation

This project is a rule-based AI recommendation system.

The system takes two inputs:

1. Interest area
2. Skill level

These inputs are passed to the `recommend_course()` function.

The function uses if/elif conditions to compare the user's preferences with predefined rules.

When a rule matches the user's preferences, the system returns the corresponding learning recommendation.

If no rule matches, the system returns a message asking the user to try another combination.

Why This Is Rule-Based AI

This system is called rule-based AI because the recommendations are determined by manually defined rules.

It does not train a Machine Learning model and does not learn from historical data.

Project Flow

User Preferences ↓ Rule Engine ↓ Rule Matching ↓ Recommendation

Rule Engine

User Interest	Skill Level	Rule	Recommendation
AI/ML	Beginner	IF interest = AI/ML AND level = Beginner	AI/ML Foundations — Beginner Track
AI/ML	Advanced	IF interest = AI/ML AND level = Advanced	AI/ML Internship — Advanced Track
Web Dev	Intermediate	IF interest = Web Dev AND level = Intermediate	Python Full Stack — Intermediate Track
Data Analysis	Beginner	IF interest = Data Analysis AND level = Beginner	Excel + Data Analysis — Beginner Track
Web Dev	Beginner	IF interest = Web Dev AND level = Beginner	Web Development — Beginner Track
Data Analysis	Advanced	IF interest = Data Analysis AND level = Advanced	Python Data Analysis — Advanced Track
Other combinations	Any	ELSE	No exact recommendation found

✓ Day 6 — Learning Outcome

I learned how to combine Python functions, conditional logic, and structured inputs to build a simple rule-based AI system.

The system takes user preferences, applies predefined rules, and produces an explainable recommendation.

The recommendation can be traced back to the specific rule that was matched.

Start coding or [generate](#) with AI.